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Date: Dec 21, 2023

No: NRSC/DMSG/2023/FL/TN/05

Dear Sir,

Sub: Tamil Nadu Heavy Rains 2023 - Near Real Time Inundation Mapping using satellite data

As you are aware, National Remote Sensing Centre is carrying out 'Near Real Time Flood Mapping' using multi-sensor satellite data. Heavy rains were reported in Tamil Nadu state in third week of Dec 2023 due to heavy rains. We have analyzed ResourceSat-2A AWIFS data on 20 Dec 2023 and a map is generated showing flood inundation / standing water in rain affected areas in coastal districts of Tamil Nadu. The map along with a brief report is enclosed herewith for your kind information.

A shape file (GIS format) of spatial inundation layer is sent to TNSDMA (Tamil Nadu State Disaster Management Authority) by e-mail for further value addition and dissemination to user departments.

Further, we request you to visit flood page in Bhuvan portal (<https://bhuvan-app1.nrsc.gov.in/disaster/disaster.php?Id=flood>) NDEM portal (<https://ndem.nrsc.gov.in>) for better visualization of flood extent in Tamil Nadu State.

We will continue to keep a watch on the situation and update the information as and when satellite data is available. You may please share the field reports received during this event for correlating the field situation during satellite image analysis. Please provide feedback on the information being provided.

With kind regards

Yours sincerely,
(A V SURESH BABU)

As per the distribution list enclosed in the report.

Mapping and Monitoring of Flood situation in Tamil Nadu State -Heavy Rains -Dec 2023

Heavy rains were witnessed during third week of December 2023 in southern parts of Tamil Nadu. Water level reported beyond danger level in Tambraparni River and there are several low lying areas which were reported to be under flood inundation. (Source: News Media, IMD, CWC)

As part of Disaster Management Support Programme of ISRO, National Remote Sensing Centre (NRSC) Flood Disaster Team is keeping a constant watch on the flood situation for enabling the acquisition and interpretation of multi-sensor satellite data for identification of flood inundation areas generation of flood inundation map. ResourceSat-2A AWIFS data of 20th Dec 2023, is acquired and analyzed (Refer Table-1).

Table-1 Satellite Data acquired over Flood Affected Area

S No	Date	Satellite/Sensor	Districts affected
1	07-12-2023 (0600 Hrs)	RISAT-1A (EOS-04) MRS SAR	Kanchipuram, Chengalpattu, Cuddalore, Ranipettai, Tiruvallur, Tiruvannamalai, Villupuram, Thiruvallur, Chennai and Vellore
2	18-12-2023 (1800 Hrs)	RISAT-1A (EOS-04) FRS SAR	Tuticorin district
3	20-12-2023 (1030 Hrs)	ResourceSat-2A AWIFS	Tuticorin district

Satellite data is geometrically rectified and flood inundation layer is extracted. The inundation layer is integrated with district boundaries layer, for estimation of district wise inundation statistics as shown Table-2. Based on the analysis of satellite data, It is observed that flood inundation exists in parts of coastal districts to an extent of 4416 Ha covering Tuticorin district.

Table-2 :District-wise inundated area statistics

S. No.	District	Inundated Area (Ha)
1	Tuticorin	4416
Total		4416

Disclaimer : This Map is prepared on rapid mapping mode for immediate use and sharing amongst official agencies. Map provides preliminary results. Flood inundation may include rain water accumulation / flood water in low lying areas. All geographic information has limitations due to the scale, resolution, date and interpretation of the original source materials. No ground verification is done.

Dissemination of Flood Maps and Reports	
Govt. of India	• Additional Secretary, MHA • Joint Secretary, DM, MHA • Control Room, MHA • Director General, NDRF, MHA • Joint Secretary , NDMA, MHA • Control Room, NDMA, MHA • Director, Flood Control Authority , CWC, MoJS • Flood Monitoring Directorate, CWC MoJS • India Meteorological Department (IMD)
ISRO/DOS	• Scientific Secretary, ISRO • Director, EDPO, ISRO • Associate Director, EDPO, ISRO • Director, NRSC • Associate Director, NRSC • Deputy Director (RSA) & PD, RRES, NRSC • Group Director, DMSG, NRSC
Tamil Nadu State Govt.	• Additional Chief secretary/ Commissioner of Revenue Administration. • Director (Disaster Management) • Joint Director (Disaster Management)